Dynamics of quasiparticles — Solution

Y23 (2023) — English translation

A1 **0.70 pts**

Let the dependence of electron energy on momentum have the form

$$\varepsilon(p) = \varepsilon_0(3 - \cos(ap_x) - \cos(ap_y) - \cos(ap_z)).$$

Find it acceleration in moment time, when momentum equal $p_x = \dfrac{\pi}{2a}$, $p_y = \dfrac{\pi}{3a}$, $p_z = 0$, *and component act on electron force* F_x, F_y, F_z.

Let's express the speed of the electron in terms of its momentum:

$$v_x = \frac{\partial \varepsilon}{\partial p_x} = \varepsilon_0 a \sin(ap_x),$$

similarly for remaining two component

$$v_y = \varepsilon_0 a \sin(ap_y),\ v_z = \varepsilon_0 a \sin(ap_z).$$

From the equation motion $\dfrac{d}{dt}\vec{p} = \vec{F}$. Express acceleration through derivative momentum:

$$a_x = \dot{v}_x = \varepsilon_0 a^2 \cos(ap_x)\dot{p}_x = \varepsilon_0 a^2 \cos(ap_x) F_x.$$

The formulas for the remaining components are similar. For impulse values from the condition
Answer:

$$a_x = 0,\ a_y = \frac{1}{2}\varepsilon_0 a^2 F_y,\ a_z = \varepsilon_0 a^2 F_z.$$

A2 **0.80 pts**

Let an electron with $\varepsilon(\vec{p})$ *from A1 move in a constant electric field of magnitude* E *directed along the* x *axis. The initial coordinates and projections of the momentum are equal to zero. Electron charge* $-q$. *Find his law of motion* $x(t)$.

The electron is acted upon by the force $F_x = -qE$, so the momentum

$$p_x = -qEt,$$

and correspond velocity

$$v_x = -a\varepsilon_0 \sin(aqEt).$$

Remaining projection velocity equal zero. Integrate, find dependence coordinate x as a function of time
Answer:

$$x(t) = -\frac{\varepsilon_0}{qE}(1 - \cos(aqEt))$$

A3 **0.50 pts**

The concentration of free electrons in a certain substance is n, *the dependence is* $\varepsilon(p)$ *as in A1. At the initial moment of time, a constant electric field* E *is turned on, directed along the axis* x. *Find the dependence of the current density* j *on time.*

Current density is expressed in terms of electron speed and concentration

$$j_x = -nqv_x = nqa\varepsilon_0 \sin(aqEt).$$

Answer:

$$j_x = nqa\varepsilon_0 \sin(aqEt).$$

B1 — 0.50 pts

Let the dependence of energy on momentum $\varepsilon(\vec{p})$ be arbitrary. Prove that the energy ε and the projection of momentum onto the direction of the magnetic field are conserved.

The equation of motion is

$$\dot{\vec{p}} = -q\vec{v} \times \vec{B}.$$

Derivative energy by time

$$\frac{d\varepsilon}{dt} = \frac{\partial \varepsilon}{\partial p_x}\dot{p}_x + \frac{\partial \varepsilon}{\partial p_y}\dot{p}_y + \frac{\partial \varepsilon}{\partial p_z}\dot{p}_z = v_x\dot{p}_x + v_y\dot{p}_y + v_z\dot{p}_z = \vec{v}\dot{\vec{p}}.$$

Substitute expression for derivative momentum, obtain

$$\dot{\varepsilon} = \vec{v} \cdot \left(-q\vec{v} \times \vec{B}\right) = 0.$$

Taking the scalar product of the equation of motion with the magnetic field, we obtain

$$\frac{d}{dt}(\vec{B}\vec{p}) = \vec{B}\dot{\vec{p}} = -q\vec{B}(\vec{v} \times \vec{B}) = 0,$$

that is, the projection of the impulse onto the magnetic field is constant.

Answer:

$$\dot{\varepsilon} = \vec{v} \cdot \left(-q\vec{v} \times \vec{B}\right) = 0.$$

B2 — 1.10 pts

Let the dependence of energy on momentum have the form

$$\varepsilon = \frac{p_x^2}{2m_x} + \frac{p_y^2}{2m_y}.$$

Energy electron equal ε. Find frequency motion ω. Depict trajectory motion particle on plane xy, indicate the characteristic dimensions and direction of movement. Place the origin of coordinates at the center of symmetry of the trajectory.

Let's write the equations of motion:

$$m_x\dot{v}_x\& = -qv_yB;$$
$$m_y\dot{v}_y\& = qv_xB.$$

Let's exclude v_y from the first equation by differentiating it and expressing $\dot{v}_y$ from the second equation. We will receive

$$m_x\ddot{v}_x = -q\dot{v}_yB = -\frac{q^2B^2}{m_y}v_x,$$

that is

$$\ddot{v}_x = -\frac{q^2B^2}{m_xm_y}v_x.$$

This is the equation of harmonic oscillations with frequency

$$\omega = \frac{qB}{\sqrt{m_xm_y}}.$$

The dependence of speed on time has the form

$$v_x = A\cos(\omega t + \varphi),$$

where amplitude A one can find, know energy electron:

$$A = \sqrt{\frac{2\varepsilon}{m_x}}.$$

Second component velocity one can find from the equation motion

$$v_y = -\frac{m_x}{qB}\dot{v}_x = \sqrt{\frac{2\varepsilon}{m_x}}\frac{\omega}{qB}\sin(\omega t + \varphi) = \sqrt{\frac{2\varepsilon}{m_y}}\sin(\omega t + \varphi).$$

Integrate, obtain trajectory particle

$$x(t)\& = \sqrt{\frac{2\varepsilon}{m_x}}\frac{1}{\omega}\sin(\omega t + \varphi) = \frac{\sqrt{2m_y\varepsilon}}{qB}\sin(\omega t + \varphi);$$
$$y(t)\& = -\frac{\sqrt{2m_x\varepsilon}}{qB}\cos(\omega t + \varphi).$$

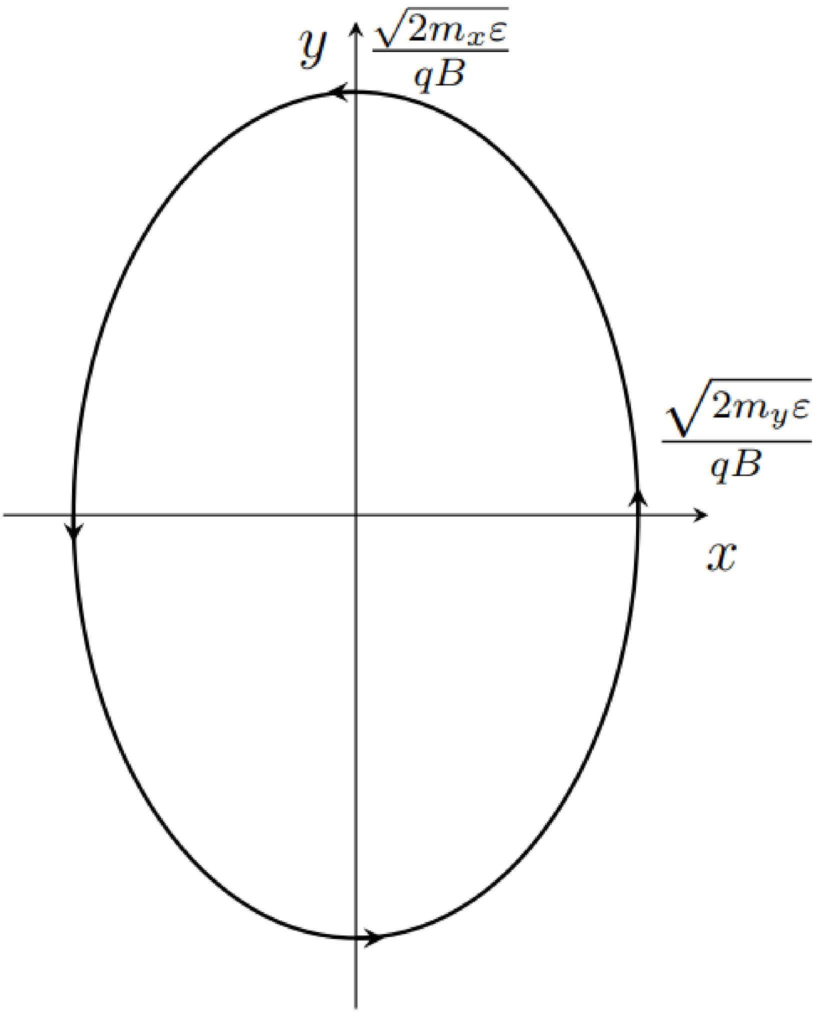
Thus, trajectory is ellipse with semi-axis

$$A_x = \frac{\sqrt{2m_y\varepsilon}}{qB},\ A_y = \frac{\sqrt{2m_x\varepsilon}}{qB}.$$

The movement occurs counterclockwise.

Answer:

$$\omega = \frac{qB}{\sqrt{m_x m_y}},\ A_x = \frac{\sqrt{2m_y\varepsilon}}{qB},\ A_y = \frac{\sqrt{2m_x\varepsilon}}{qB}.$$



B3 **0.60 pts**

Find the hodograph of the electron's momentum vector (the trajectory along which the end of the momentum vector moves if it is plotted from one point). Draw it on the plane p_x, p_y. Please indicate typical dimensions.

Taking into account the electric field, the equations of motion are

$$m_x\dot{v}_x\& = -qE - qv_yB;$$
$$m_y\dot{v}_y\& = qv_xB.$$

Let us introduce a new variable u_y using the relation

$$v_y = -\frac{E}{B} + u_y,$$

then equations of motion assume such same form, as and in previous part:

$$m_x\dot{v}_x\& = -qu_yB;$$
$$m_y\dot{u}_y\& = qv_xB.$$

Initial condition

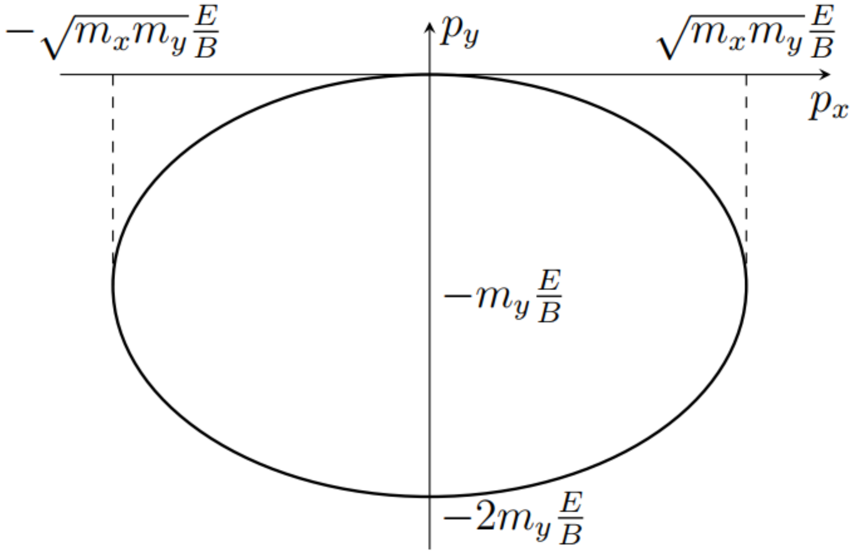
$$v_x(0) = 0,\ v_y(0) = 0,\ u_y(0) = \frac{E}{B}.$$

Taking into account this solution

$$v_x = -\sqrt{\frac{m_y}{m_x}}\frac{E}{B}\sin\omega t;\ v_y = -\frac{E}{B}(1-\cos\omega t).$$

Then momentum

$$p_x = -\sqrt{m_x m_y}\frac{E}{B}\sin\omega t,\ p_y = -m_y\frac{E}{B}(1-\cos\omega t).$$



B4

0.40 pts

Find the law of motion $x(t)$, $y(t)$.

Integrating the speeds from the previous point, we get

Answer:

$$x(t)\& = -\frac{m_y E}{qB^2}(1-\cos\omega t)$$

$$y(t)\& = -\frac{E}{B}t + \frac{\sqrt{m_x m_y}E}{qB^2}\sin\omega t,\ \omega = \frac{qB}{\sqrt{m_x m_y}}.$$

B5

1.40 pts

If the energy is greater than the minimum value, the trajectories on the plane become open, and the particle can travel indefinitely in some direction. Find this minimum energy ε_{min}. At energy $\varepsilon > \varepsilon_{min}$, draw the trajectory and indicate the characteristic geometric dimensions.

Let us write down the equations of motion

$$\dot{\vec{p}} = -q\vec{v}\times\vec{B} = -q\dot{\vec{r}}\times\vec{B}.$$

Integrate, obtain

$$\Delta\vec{p} = -q\Delta\vec{r}\times\vec{B}.$$

Multiply this equality vector on $\vec{B}$.

$$\Delta\vec{p}\times\vec{B} = -q(\Delta\vec{r}\times\vec{B})\times\vec{B} = qB^2\Delta\vec{r}.$$

Thus, vector displacement one can express through change momentum

$$\Delta\vec{r} = \frac{1}{qB^2}\Delta\vec{p}\times\vec{B}.$$

Let us first consider it in momentum space. From energy conservation it follows that the momentum hodograph is the surface

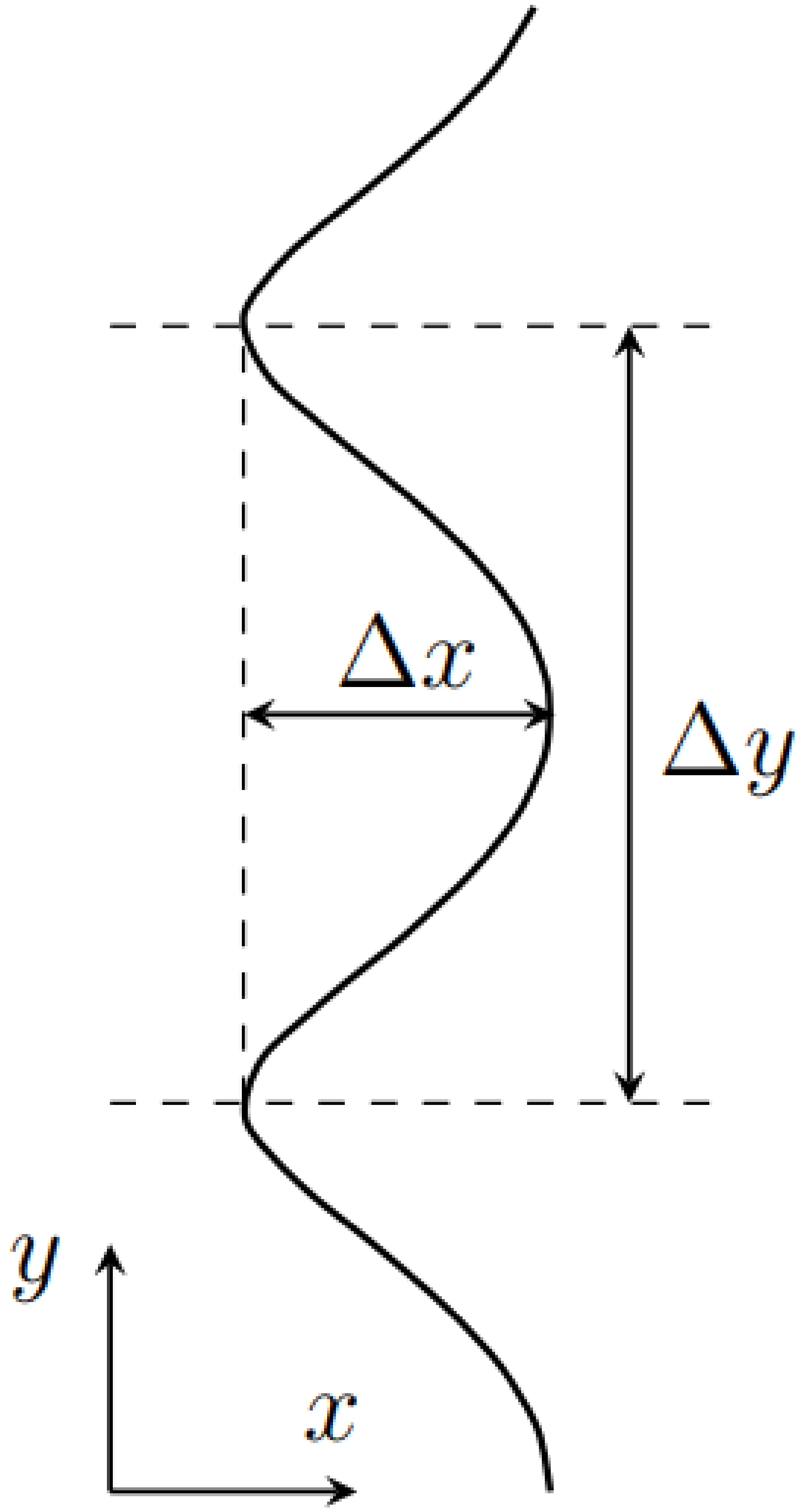
$$\frac{p_y^2}{2m_y} + \varepsilon_0(1-\cos ap_x) = \varepsilon = const.$$

Equation this surface one can represent in form

$$p_y = \pm\sqrt{2m_y(\varepsilon - \varepsilon_0 + \varepsilon_0 \cos ap_x)}.$$

in that case, if $\varepsilon \leq 2\varepsilon_0$, hodograph is a closed curve. Then the trajectory is also a closed curve. If however $\varepsilon > 2\varepsilon_0$, hodograph the momentum is an unclosed curve directed along the x axis. Then the trajectory motion particle - curve, direct along y, moreover dependence $x(y)$ – periodic. Find the period of this function and the width of the strip along the x axis that it occupies.

$$\Delta y = \frac{1}{qB}\frac{2\pi}{a}, \ \Delta x = \frac{1}{qB}\Delta p_y = \frac{\sqrt{2m_y}}{qB}(\sqrt{\varepsilon} - \sqrt{\varepsilon - 2\varepsilon_0}).$$



C1 **0.50 pts**

First, we neglect the variance and consider the case $\alpha = 0$. At what angles θ_1, θ_2 is decay possible?

Let's write down the laws of conservation of energy and momentum:

$$up = uq_1 + uq_2,\ \vec{p} = \vec{q}_1 + \vec{q}_2.$$

Since length vector $\vec{p}$ is equal to the sum of the lengths of the remaining two vectors, all three vectors must be directed in the same direction.

Answer: $\theta_1 = \theta_2 = 0$

C2 — 0.50 pts

Express the modulus of the second phonon q_2 in terms of p, q_1, θ_1. Obtain the expression in the limit of small θ_1 up to second order in θ_1.

From the law of conservation of momentum

$$\vec{q}_2 = \vec{p} - \vec{q}_1.$$

Hence

$$q_2^2 = p^2 + q_1^2 - 2pq_1\cos\theta_1 = p^2 + q_1^2 - 2pq_1(1 - \theta_1^2/2) = (p - q_1)^2 + pq_1\theta_1^2.$$

We extract the root and expand in a series to the required order

$$q_2 = \sqrt{(p - q_1)^2 + pq_1\theta_1^2} \approx p - q_1 + \frac{pq_1}{2(p - q_1)}\theta_1^2$$

C3 — 1.20 pts

The angle dependence of the emitted phonon momentum at small angles has the form

$$q_1(\theta_1) = A + B\theta_1.$$

Determine constant A, B. Express they through p, u, α.

Let us write the law of conservation of energy

$$up + \alpha p^3 = uq_1 + \alpha q_1^3 + uq_2 + \alpha q_2^3.$$

Substitute expression for q_2:

$$up + \alpha p^3 = uq_1 + \alpha q_1^3 + u\left(p - q_1 + \frac{pq_1}{2(p - q_1)}\theta_1^2\right) + \alpha\left(p - q_1 + \frac{pq_1}{2(p - q_1)}\theta_1^2\right)^3.$$

Let us expand the brackets and keep contributions of order θ_1^2:

$$\alpha p^3 = \alpha q^3 + \frac{upq_1}{2(p - q_1)}\theta_1^2 + \alpha(p - q_1)^3 + \frac{3\alpha}{2}pq_1(p - q_1)\theta_1^2.$$

Move the term that does not depend on the angle to the left:

$$\alpha(p^3 - q_1^3 - (p - q_1)^3) = \frac{upq_1\theta_1^2}{2(p - q_1)}\left(1 + \frac{3\alpha}{u}(p - q_1)^2\right).$$

Second term in bracket in right part one can neglect in force condition $\alpha p^3 \ll up$, on the left side we expand the bracket and obtain

$$3\alpha(p - q_1)pq_1 = \frac{upq_1\theta_1^2}{2(p - q_1)}.$$

Hence

$$(p - q_1)^2 = \frac{u}{6\alpha}\theta_1^2,\ q_1 = p - \sqrt{\frac{u}{6\alpha}}\theta_1.$$

Answer:

$$A = p,\ B = -\sqrt{\frac{u}{6\alpha}}.$$

C4 — 0.30 pts

At what sign of the constant α is decay possible?

The equation in the previous paragraph has a solution if $\alpha > 0$.

Answer: $\alpha > 0$

C5 — 0.50 pts

Find the maximum angle θ_{max} at which a phonon can be emitted. Under what condition on p can the phonon momentum be considered small?

The maximum angle is achieved at the minimum momentum of the emitted phonon $q_1 \to 0$. Then from the expression for q_1 we find

$$\theta_{\max} = \sqrt{\frac{6\alpha}{u}} p.$$

The whole problem was solved in the small-angle approximation. This approximation works under the condition $\theta_{max} \ll 1$, that is

$$p \ll \sqrt{\frac{u}{\alpha}}.$$

Answer:

$$\theta_{\max} = \sqrt{\frac{6\alpha}{u}} p.$$

C6 — 1.00 pts

Express the angle θ_2 at which the second phonon is emitted in terms of θ_1, p, u, α. Consider the angles small, take into account the first-order contributions from θ_1, θ_2.

From the results C3 it follows that the phonon momenta are expressed in terms of emission angles as

$$q_1 = p - \sqrt{\frac{u}{6\alpha}} \theta_1, \ q_2 = p - \sqrt{\frac{u}{6\alpha}} \theta_2.$$

Write projection law conservation momentum on the axis perpendicular to the direction of motion of the original phonon:

$$q_1\theta_1 \& = q_2\theta_2,$$

$$p\theta_1 - \sqrt{\frac{u}{6\alpha}} \theta_1^2 \& = p\theta_2 - \sqrt{\frac{u}{6\alpha}} \theta_2^2.$$

Let us regroup the terms:

$$p(\theta_1 - \theta_2) = \sqrt{\frac{u}{6\alpha}} (\theta_1^2 - \theta_2^2), \ p = \sqrt{\frac{u}{6\alpha}} (\theta_1 + \theta_2).$$

Answer:

$$\theta_2 = \sqrt{\frac{6\alpha}{u}} p - \theta_1$$